

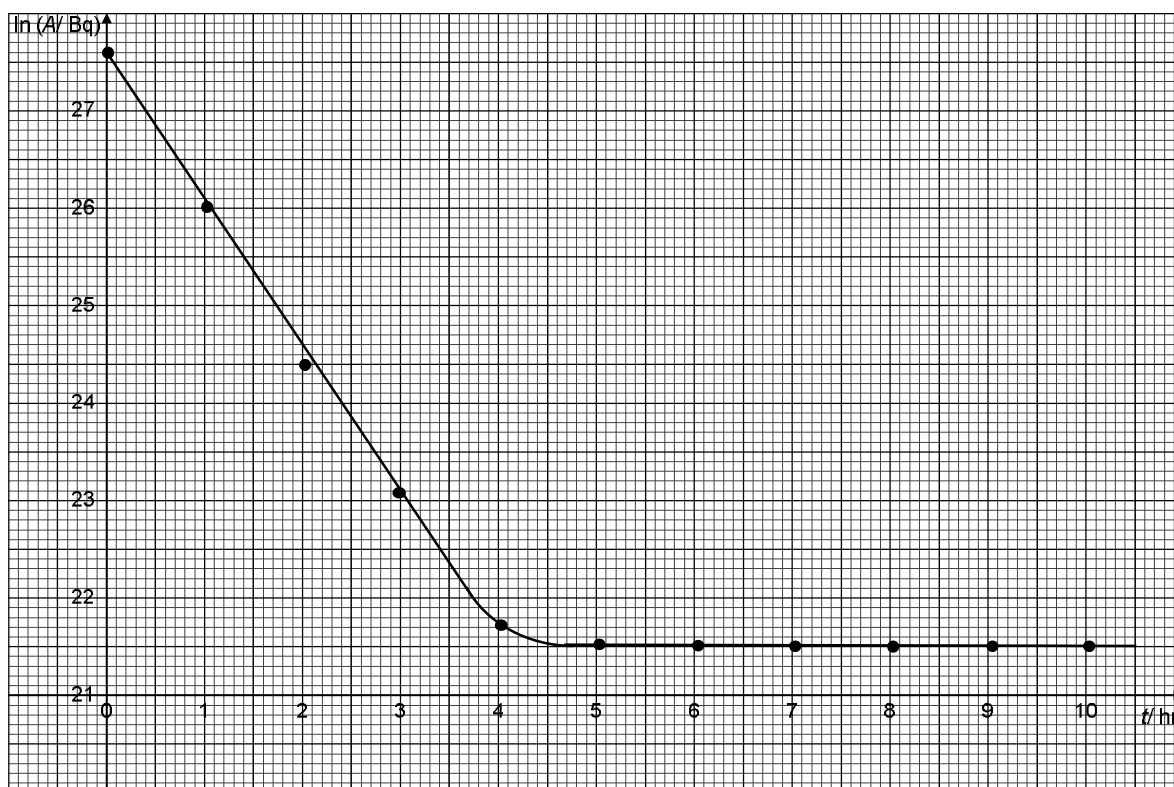
- 6(a) Spontaneous means that the decay process occurs on its own accord and are unaffected by physical or chemical means.

Half-life of a radioactive element is the *average* time taken for half the original number of the radioactive nuclei to decay.

- (b)(i) From the table, at $t = 3$ hr, $A = 1.08 \times 10^{10} \text{ Bq}$

$$\Rightarrow \ln A = \ln(1.08 \times 10^{10}) = 23.103$$

(ii)



- (iii) From 0 to 3 hrs, the graph is linear with a negative slope. This is because the activity is predominantly that of $^{131}_{52}\text{Te}$ as there are only small amounts of $^{131}_{53}\text{I}$. So the combined activity follows the linear equation: $\ln A = -\lambda t + \ln A_0$ for $^{131}_{52}\text{Te}$.

From 3 to 6 hrs, the graph is a curve as the activity of $^{131}_{53}\text{I}$ nuclei grows and the activity of $^{131}_{52}\text{Te}$ decreases. So the resultant activity is due to two different nuclides, and doesn't follow the linear equation above.

From 6 to 10 hrs, the line is horizontal due mainly to the activity of $^{131}_{53}\text{I}$ which has a very long half-life as compared to $^{131}_{52}\text{Te}$. A long half-life means a low decay rate, which means that the activity hardly changes.

- (iv) Decay constant of $^{131}_{52}\text{Te}$ can be obtained from the *gradient* of the linear part of the graph from 0 to 3 hrs.

$$\therefore \lambda = -\frac{27.60 - 23.40}{0.00 - 2.80} = 1.50 \text{ hr}^{-1}$$

- (v) From the table, at $t = 0$ hr, $A_0 = 1.00 \times 10^{12} \text{ Bq}$

Since, $A = \lambda N$

$$\Rightarrow N_0 = \frac{A_0}{\lambda} = \frac{1.00 \times 10^{12}}{1.5 / (60 \times 60)} = 2.40 \times 10^{15}$$

- (vi) Initially the rate of production of $^{131}_{53}\text{I}$ is much *higher* than the rate of its decay to $^{131}_{54}\text{Xe}$, hence the amount of $^{131}_{53}\text{I}$ increases.

Later, most of the $^{131}_{52}\text{Te}$ would have decayed, resulting in the rate of production of $^{131}_{53}\text{I}$ to be *lower* than its rate of decay to $^{131}_{54}\text{Xe}$. Hence, the amount of $^{131}_{53}\text{I}$ will then decrease.

- (vii) Assuming that all the $^{131}_{52}\text{Te}$ has been converted to $^{131}_{53}\text{I}$, the number of $^{131}_{53}\text{I}$ isotopes after 6 hours is 2.40×10^{15} and its activity is 2.12×10^9 Bq (*from table*).

$$\text{Since, } A = \lambda N = \frac{\ln 2}{t_{1/2}} N$$

$$\Rightarrow t_{1/2} = \frac{\ln 2}{A} N = \frac{\ln 2}{2.12 \times 10^9} (2.40 \times 10^{15}) = 7.85 \times 10^5 \text{ s}$$

7(a) $\lambda = \nu / f$